

Qualification Pack



Green Hydrogen Plant Technician

QP Code: SGJ/Q0120

Version: 1.0

NSQF Level: 4

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SGJ/Q0120: Green Hydrogen Plant Technician

Brief Job Description

Green Hydrogen Plant Technician shall perform testing, installation and facility integration of electrolyser, repairs, troubleshooting, upkeep and maintenance of electrical control systems, protection systems, and other auxiliary equipment and associated tools in Green hydrogen generation facilities. The job holder will be responsible for the continuous supply of renewable power for its integration with the electrolyzers and the other process equipment for generating Green Hydrogen. This role works closely with the power supply project, testing, plant engineering, process operation, control & operation across the green hydrogen generation plant.

Personal Attributes

This job requires the individual to be sincere, diligent, hardworking and attentive to effectively perform the work timely in a quality manner while following all the safety practices at Green Hydrogen Manufacturing facility.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [SGJ/N0150: Introduce green hydrogen for clean energy transition](#)
2. [SGJ/N4012: Identify components of Green Hydrogen Plant and interpret its layout](#)
3. [SGJ/N0156: Integrate renewable power sources for Green Hydrogen Plant](#)
4. [SGJ/N0155: Perform installation of electrolyzer for green hydrogen generation](#)
5. [SGJ/N0154: Analyze input water system to electrolyzer for green hydrogen generation](#)
6. [SGJ/N0153: Operate Hydrogen conditioning and compression unit](#)
7. [SGJ/N0152: Set up and operate Hydrogen storage system](#)
8. [SGJ/N0151: Commissioning of Green Hydrogen system](#)
9. [SGJ/N0802: Maintain Health & Safety at Green Hydrogen generation project site](#)
10. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)

Qualification Pack (QP) Parameters

Qualification Pack

Sector	Green Jobs
Sub-Sector	Renewable Energy
Occupation	Hydrogen Plant Technician
Country	India
NSQF Level	4
Credits	14
Aligned to NCO/ISCO/ISIC Code	NCO-2015/ 8131.2100 Ammonia Operator/Ammonia Plant Operator
Minimum Educational Qualification & Experience	12th Class with NA of experience OR Completed 2nd year of the 3-year diploma after 10 (and pursuing diploma) OR 10th grade pass (plus 2-year NTC) OR 10th grade pass with 1 year NTC plus NAC OR 11th grade pass (and pursuing continuous schooling) OR 10th grade pass with 2 Years of experience relevant OR Previous relevant Qualification of NSQF Level (NSQF Level 3.0 with minimum education as 8th Grade pass) with 3 Years of experience relevant
Minimum Level of Education for Training in School	Not Applicable
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	30/06/2026
NSQF Approval Date	31/01/2023
Version	1.0
Reference code on NQR	QG-04-ES-00146-2023-V1-SCGJ
NQR Version	1



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Remarks:

Total 420 Hours or 14 Credits : (Theory: 160 hours+Practical:140 hours+ 60 hours of employability skills + 60 hours of OJT)

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SGJ/N0150: Introduce green hydrogen for clean energy transition

Description

This unit outlines the basics of Green Hydrogen technology and the evolving hydrogen ecosystem as a key pathway for energy transition

Scope

The scope covers the following :

- Fundamental concepts of Green Hydrogen
- Skills Required for the Green Hydrogen Economy

Elements and Performance Criteria

Fundamental concepts of Green Hydrogen

To be competent, the user/individual on the job must be able to:

- PC1.** Explain the necessity of green hydrogen in sustainable energy transition
- PC2.** Discuss the properties and characteristics of Hydrogen
- PC3.** Describe basic concepts of Hydrogen as an energy carrier
- PC4.** Demonstrate with chart colour code nomenclature of Hydrogen
- PC5.** Perform an activity for matching the process and source of production as applicable for different colour codes of hydrogen
- PC6.** Discuss the value chain for Green Hydrogen
- PC7.** Discuss briefly various technology options for production of green hydrogen and outline benefits and drawbacks of existing methods of Hydrogen production
- PC8.** Briefly discuss the end use applications of Green hydrogen in industry, transport and power production.
- PC9.** Draw a flow diagram of green hydrogen production, conversion and end uses across the energy system
- PC10.** Discuss key aspects and challenges related to production, storage, transportation, distribution of Green Hydrogen.
- PC11.** Discuss various renewable energy sources for large scale production of green hydrogen through electrolysis of water

Skills Required for the Green Hydrogen Economy

To be competent, the user/individual on the job must be able to:

- PC12.** Discuss Green hydrogen economy in Indian context along with the key aspects of Green Hydrogen Policy and Mission of Government of India
- PC13.** Show how green hydrogen economy is critical for energy transition and the key policy measures announced by the Government of India
- PC14.** Illustrate key announcements made by the leading private and public sector companies across the value chain of Green hydrogen and outline how those would create new opportunities for technicians.
- PC15.** Discuss the role and responsibilities of the job holder

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Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** properties of hydrogen
- KU2.** fundamental differences between hydrogen and other fuels
- KU3.** specific requirements relating to hydrogen infrastructure
- KU4.** Understanding relevant regulations and safety standards
- KU5.** Hydrogen application across sectors

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read vernacular/English language
- GS2.** participate in and understand the main points of discussions with teams and external stakeholders
- GS3.** express statements or information clearly so that others can understand
- GS4.** plan and organize work schedule to meet work deadlines
- GS5.** take decision with systematic course of actions and/or response
- GS6.** read and write different types of documents/instructions/correspondence
- GS7.** communicate effectively using appropriate language in formal and informal settings
- GS8.** behave politely and appropriately with all stakeholder

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Fundamental concepts of Green Hydrogen</i>	20	13	-	-
PC1. Explain the necessity of green hydrogen in sustainable energy transition	2	-	-	-
PC2. Discuss the properties and characteristics of Hydrogen	2	-	-	-
PC3. Describe basic concepts of Hydrogen as an energy carrier	2	-	-	-
PC4. Demonstrate with chart colour code nomenclature of Hydrogen	-	3	-	-
PC5. Perform an activity for matching the process and source of production as applicable for different colour codes of hydrogen	-	4	-	-
PC6. Discuss the value chain for Green Hydrogen	3	-	-	-
PC7. Discuss briefly various technology options for production of green hydrogen and outline benefits and drawbacks of existing methods of Hydrogen production	3	3	-	-
PC8. Briefly discuss the end use applications of Green hydrogen in industry, transport and power production.	3	-	-	-
PC9. Draw a flow diagram of green hydrogen production, conversion and end uses across the energy system	-	3	-	-
PC10. Discuss key aspects and challenges related to production, storage, transportation, distribution of Green Hydrogen.	3	-	-	-
PC11. Discuss various renewable energy sources for large scale production of green hydrogen through electrolysis of water	2	-	-	-
<i>Skills Required for the Green Hydrogen Economy</i>	10	7	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. Discuss Green hydrogen economy in Indian context along with the key aspects of Green Hydrogen Policy and Mission of Government of India	3	-	-	-
PC13. Show how green hydrogen economy is critical for energy transition and the key policy measures announced by the Government of India	2	4	-	-
PC14. Illustrate key announcements made by the leading private and public sector companies across the value chain of Green hydrogen and outline how those would create new opportunities for technicians.	3	3	-	-
PC15. Discuss the role and responsibilities of the job holder	2	-	-	-
NOS Total	30	20	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N0150
NOS Name	Introduce green hydrogen for clean energy transition
Sector	Green Jobs
Sub-Sector	Renewable Energy
Occupation	Installation, Operation and Maintenance
NSQF Level	4
Credits	1
Version	1.0
Last Reviewed Date	NA
Next Review Date	31/01/2026
NSQC Clearance Date	31/01/2023

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SGJ/N4012: Identify components of Green Hydrogen Plant and interpret its layout

Description

This unit outlines aspects on key components of a Green Hydrogen generation system

Scope

The scope covers the following :

- Key Green Hydrogen system components
- Plant Layout

Elements and Performance Criteria

Key Green Hydrogen system components

To be competent, the user/individual on the job must be able to:

- PC1.** Identify all key components of the Green Hydrogen plant including electrical, mechanical and civil components
- PC2.** Illustrate the schematic of Green hydrogen production plant
- PC3.** Discuss functions of the key components including of electrolyzer stacks, renewable power supply system, , feed water supply and demineralisation system, gas separator, transformer and rectifier, gas compression unit, etc
- PC4.** Illustrate key components of the plant and outline their functions through plant schematic
- PC5.** Discuss fundamental principles of main components on which they operate (e.g. electrolyzer stack, gas collector, power source, etc.)
- PC6.** Discuss briefly on types of electrolyzers
- PC7.** Illustrate possible combination of various renewable power sources for generating green hydrogen

Plant Layout

To be competent, the user/individual on the job must be able to:

- PC8.** Explain overall layout of the plant
- PC9.** Illustrate how to interpret the Plant Layout including various equipment and material used in a Green hydrogen production facility.
- PC10.** Discuss Sparking items e.g. MCBs, switches to be installed isolated from the hydrogen production area
- PC11.** Explain how to read and interpret electrical and other applicable codes, standards and protocols relevant to the plant.
- PC12.** Explain key material and safety codes, Technology protocols and standards as applicable in Green hydrogen industry
- PC13.** Demonstrate how to interpret signs, notices and/or cautions at project site.
- PC14.** Identify processes where energy/electricity and material utilization can be optimized

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Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** knowhow of tools & tackles required to carry out the work
- KU2.** relevant standards and regulations to be followed across hydrogen industry
- KU3.** relevant personal protective equipments used across hydrogen industry
- KU4.** document information using appropriate corporate forms

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** follow organisation's code of conduct
- GS2.** follow organization's rule-based decision making process
- GS3.** record data as per specified process
- GS4.** read from different sources- books, screens in machines and signage
- GS5.** make timely decisions for efficient utilization of resources

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Key Green Hydrogen system components</i>	16	12	-	-
PC1. Identify all key components of the Green Hydrogen plant including electrical, mechanical and civil components	3	2	-	-
PC2. Illustrate the schematic of Green hydrogen production plant	1	3	-	-
PC3. Discuss functions of the key components including of electrolyzer stacks, renewable power supply system, , feed water supply and demineralisation system, gas separator, transformer and rectifier, gas compression unit, etc	3	2	-	-
PC4. Illustrate key components of the plant and outline their functions through plant schematic	2	2	-	-
PC5. Discuss fundamental principles of main components on which they operate (e.g. electrolyzer stack, gas collector, power source, etc.)	3	-	-	-
PC6. Discuss briefly on types of electrolyzers	4	-	-	-
PC7. Illustrate possible combination of various renewable power sources for generating green hydrogen	-	3	-	-
<i>Plant Layout</i>	14	8	-	-
PC8. Explain overall layout of the plant	3	2	-	-
PC9. Illustrate how to interpret the Plant Layout including various equipment and material used in a Green hydrogen production facility.	-	2	-	-
PC10. Discuss Sparking items e.g. MCBs, switches to be installed isolated from the hydrogen production area	2	-	-	-
PC11. Explain how to read and interpret electrical and other applicable codes, standards and protocols relevant to the plant.	3	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. Explain key material and safety codes, Technology protocols and standards as applicable in Green hydrogen industry	3	-	-	-
PC13. Demonstrate how to interpret signs, notices and/or cautions at project site.	2	3	-	-
PC14. Identify processes where energy/electricity and material utilization can be optimized	1	1	-	-
NOS Total	30	20	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N4012
NOS Name	Identify components of Green Hydrogen Plant and interpret its layout
Sector	Green Jobs
Sub-Sector	Renewable Energy
Occupation	Technician
NSQF Level	4
Credits	1
Version	1.0
Last Reviewed Date	NA
Next Review Date	31/01/2026
NSQC Clearance Date	31/01/2023

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SGJ/N0156: Integrate renewable power sources for Green Hydrogen Plant

Description

This unit outlines the key aspects for integrating renewable energy for powering Electrolysers and other components of Green Hydrogen generation system

Scope

The scope covers the following :

- Flexible system operation

Elements and Performance Criteria

Flexible System Operation

To be competent, the user/individual on the job must be able to:

- PC1.** Discuss the key parameters of renewable and hybrid power plants
- PC2.** Explain how cost effective and reliable Renewable Power can be assured for the production of Green Hydrogen
- PC3.** Illustrate how reliable renewable power can be supplied through integrating various renewable energy sources for generating green hydrogen
- PC4.** Discuss the flexible system operation for the supply of power from different sources
- PC5.** Show how integration of variable renewable energy (VRE) for powering electrolyzer are performed
- PC6.** Discuss how Sizing of renewable power plant along with storage capacity is critical to meet hydrogen load demand
- PC7.** Show how to calculation of hydrogen production equipment load
- PC8.** Discuss functions of transformers and rectifiers
- PC9.** Show how rectifiers operate to cater the load of electrolyzer
- PC10.** Explain and illustrate key aspects of Maintaining stability of power supply for the green hydrogen plant, switching of number of electrolyser cells as per availability of electric power, etc
- PC11.** Illustrate the process of splitting water into hydrogen and oxygen using renewable electricity

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Properties of gas
- KU2.** use of the tools and tackles
- KU3.** Renewable system integration
- KU4.** legislative, organizational, site requirements and procedures
- KU5.** how to obtain required authorization

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- KU6.** precautions to be taken while handling different electrical and mechanical components
- KU7.** importance of wearing protective clothing and other safety gear while carrying out installation and O&M activities (PPEs)
- KU8.** usage and handling procedure of electrical components

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** complete documentation work applicable to the role
- GS2.** read various color codes, as per standard electrical, mechanical and civil nomenclature
- GS3.** follow organisational code of conduct
- GS4.** approach relevant authority when required
- GS5.** make timely decisions for efficient utilization of resources
- GS6.** record data
- GS7.** read health and safety instructions and signage

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Flexible System Operation</i>	24	26	-	-
PC1. Discuss the key parameters of renewable and hybrid power plants	3	-	-	-
PC2. Explain how cost effective and reliable Renewable Power can be assured for the production of Green Hydrogen	4	-	-	-
PC3. Illustrate how reliable renewable power can be supplied through integrating various renewable energy sources for generating green hydrogen	-	4	-	-
PC4. Discuss the flexible system operation for the supply of power from different sources	4	1	-	-
PC5. Show how integration of variable renewable energy (VRE) for powering electrolyzer are performed	1	3	-	-
PC6. Discuss how Sizing of renewable power plant along with storage capacity is critical to meet hydrogen load demand	3	3	-	-
PC7. Show how to calculation of hydrogen production equipment load	1	4	-	-
PC8. Discuss functions of transformers and rectifiers	2	1	-	-
PC9. Show how rectifiers operate to cater the load of electrolyzer	2	4	-	-
PC10. Explain and illustrate key aspects of Maintaining stability of power supply for the green hydrogen plant, switching of number of electrolyser cells as per availability of electric power, etc	3	3	-	-
PC11. Illustrate the process of splitting water into hydrogen and oxygen using renewable electricity	1	3	-	-
NOS Total	24	26	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N0156
NOS Name	Integrate renewable power sources for Green Hydrogen Plant
Sector	Green Jobs
Sub-Sector	Renewable Energy
Occupation	Installation
NSQF Level	4
Credits	1
Version	1.0
Last Reviewed Date	NA
Next Review Date	31/01/2026
NSQC Clearance Date	31/01/2023

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SGJ/N0155: Perform installation of electrolyzer for green hydrogen generation

Description

This unit outlines how to perform installation of electrolyzer for green hydrogen generation

Scope

The scope covers the following :

- Installation of Electrolyzers- I
- Installation of Electrolyzers- II

Elements and Performance Criteria

Installation of Electrolyzers -I

To be competent, the user/individual on the job must be able to:

- PC1.** Discuss the types of Electrolyzer including Polymer Electrolyte Membrane (PEM), Alkaline electrolysis (AE), Solid Oxide Electrolysis (SOEC), Anion Exchange Membranes (AEM) and the status of their technology maturity
- PC2.** Explain the key technical specifications of various Electrolyzer types (PEM, AE AEM, SOEC) including capacity and sizing
- PC3.** Illustrate different types of electrolyzer and their key specifications and technical parameters through Pictures, videos, product data sheet etc
- PC4.** Provide an overview of various electrolyzer along with their key features and comparison
- PC5.** Outline differences in PEM, AE, AEM and SOEC Electrolyzer and illustrate their schematics
- PC6.** Explain the major components of an Electrolyzer
- PC7.** Discuss the Operating principles of PEM AE and SOEC electrolyzer
- PC8.** Discuss and show how to Identify suitable tools and equipment required for Installation of electrolyzer, Plant & Machineries conforming to relevant technical sheets, safety and technical standards for proper execution of work
- PC9.** Explain the inputs / outputs of an Electrolyzer system
- PC10.** Explain key O&M requirements for electrolyzer
- PC11.** Discuss basic Technical specifications of electrolyser of each cell
- PC12.** Discuss and Illustrate how to perform calculation for estimating losses and equipment efficiency

Installation of Electrolyzers- II

To be competent, the user/individual on the job must be able to:

- PC13.** Discuss how to read and interpret drawings, Mechanical and electrical diagrams along with Plant specifications
- PC14.** Explain how to install the plant using design drawings and documents provided
- PC15.** Show how to handle different tools, equipment as per concerned standard and industry practices

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- PC16.** Show how to implement the procedure for electrolyzer stack assembly and operations
- PC17.** Discuss in detail the input renewable power from various sources and its integration with electrolyzer
- PC18.** Discuss the assembly/Installation of Parts and Components of Electrolyzer
- PC19.** Show how to perform requisite data and document management
- PC20.** Discuss step by step process for assembly/Installation of Electrolyzer- Bipolar plate, separator(membrane), Porous Transport layer, electrodes etc.
- PC21.** Show how to get familiar with mechanical/electrical systems for system installation, troubleshooting and commissioning work
- PC22.** Discuss how to perform stack assembling, testing, operation, maintaining and troubleshooting of an electrolyzer
- PC23.** Show how to ensure that design guidelines and relevant engineering codes and standards are rigorously followed for installation, commission and maintenance of the electrolyzer
- PC24.** Discuss how to perform stack assembling, testing, operation, maintaining and troubleshooting of an electrolyzer
- PC25.** Show how to perform assembling, testing, maintaining and troubleshooting of an electrolyzer
- PC26.** Discuss Key selection parameters of electrolysis including density, pressure, operating temperature, hydrogen purity, Export component and Volume and weight
- PC27.** Discuss the importance of Manifold and sealing.
- PC28.** Explain how to Maintain the log for daily operation of electrolyzer
- PC29.** Explain Start up, shutdown, and operate the production processes
- PC30.** Show how to monitor various parameters of electrolysis
- PC31.** Demonstrate how to support in performing process controls functions in conjunction with overall project engineering and other functional teams

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** legislative, organizational, site requirements and related procedures
- KU2.** usage and handling procedure of system components
- KU3.** different mechanical and electrical equipment, their functioning and precautions
- KU4.** importance of wearing protective clothing and other safety gear
- KU5.** tools involved in installation and O&M activities
- KU6.** importance of safety drills
- KU7.** reporting procedures
- KU8.** cleaning procedures for various equipment

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** undertake complete documentation



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- GS2.** correctly read and interpret various color codes, as per standard electrical, mechanical and civil nomenclature
- GS3.** follow organizational rule-based decision making process
- GS4.** approach relevant authority when required
- GS5.** work constructively and collaboratively with others
- GS6.** express statements with required clarity
- GS7.** make timely decisions for efficient utilization of resources

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Installation of Electrolyzers -I</i>	30	20	-	-
PC1. Discuss the types of Electrolyzer including Polymer Electrolyte Membrane (PEM), Alkaline electrolysis (AE), Solid Oxide Electrolysis (SOEC), Anion Exchange Membranes (AEM) and the status of their technology maturity	2	2	-	-
PC2. Explain the key technical specifications of various Electrolyzer types (PEM, AE AEM, SOEC) including capacity and sizing	4	-	-	-
PC3. Illustrate different types of electrolyzer and their key specifications and technical parameters through Pictures, videos, product data sheet etc	1	4	-	-
PC4. Provide an overview of various electrolyzer along with their key features and comparison	2	2	-	-
PC5. Outline differences in PEM, AE, AEM and SOEC Electrolyzer and illustrate their schematics	2	3	-	-
PC6. Explain the major components of an Electrolyzer	2	-	-	-
PC7. Discuss the Operating principles of PEM AE and SOEC electrolyzer	2	-	-	-
PC8. Discuss and show how to Identify suitable tools and equipment required for Installation of electrolyzer, Plant & Machineries conforming to relevant technical sheets, safety and technical standards for proper execution of work	3	4	-	-
PC9. Explain the inputs / outputs of an Electrolyzer system	3	1	-	-
PC10. Explain key O&M requirements for electrolyzer	3	-	-	-
PC11. Discuss basic Technical specifications of electrolyser of each cell	3	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. Discuss and Illustrate how to perform calculation for estimating losses and equipment efficiency	3	4	-	-
<i>Installation of Electrolyzers- II</i>	20	30	-	-
PC13. Discuss how to read and interpret drawings, Mechanical and electrical diagrams along with Plant specifications	2	-	-	-
PC14. Explain how to install the plant using design drawings and documents provided	2	-	-	-
PC15. Show how to handle different tools, equipment as per concerned standard and industry practices	-	2	-	-
PC16. Show how to implement the procedure for electrolyzer stack assembly and operations	-	2	-	-
PC17. Discuss in detail the input renewable power from various sources and its integration with electrolyzer	2	3	-	-
PC18. Discuss the assembly/Installation of Parts and Components of Electrolyzer	2	-	-	-
PC19. Show how to perform requisite data and document management	-	2	-	-
PC20. Discuss step by step process for assembly/Installation of Electrolyzer- Bipolar plate, separator(membrane), Porous Transport layer, electrodes etc.	2	-	-	-
PC21. Show how to get familiar with mechanical/electrical systems for system installation, troubleshooting and commissioning work	-	3	-	-
PC22. Discuss how to perform stack assembling, testing, operation, maintaining and troubleshooting of an electrolyzer	2	2	-	-
PC23. Show how to ensure that design guidelines and relevant engineering codes and standards are rigorously followed for installation, commission and maintenance of the electrolyzer	-	3	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC24. Discuss how to perform stack assembling, testing, operation, maintaining and troubleshooting of an electrolyzer	2	-	-	-
PC25. Show how to perform assembling, testing, maintaining and troubleshooting of an electrolyzer	-	3	-	-
PC26. Discuss Key selection parameters of electrolysis including density, pressure, operating temperature, hydrogen purity, Export component and Volume and weight	2	-	-	-
PC27. Discuss the importance of Manifold and sealing.	2	-	-	-
PC28. Explain how to Maintain the log for daily operation of electrolyzer	-	1	-	-
PC29. Explain Start up, shutdown, and operate the production processes	2	3	-	-
PC30. Show how to monitor various parameters of electrolysis	-	3	-	-
PC31. Demonstrate how to support in performing process controls functions in conjunction with overall project engineering and other functional teams	-	3	-	-
NOS Total	50	50	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N0155
NOS Name	Perform installation of electrolyzer for green hydrogen generation
Sector	Green Jobs
Sub-Sector	Renewable Energy
Occupation	Installation
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	NA
Next Review Date	31/01/2026
NSQC Clearance Date	31/01/2023

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SGJ/N0154: Analyze input water system to electrolyzer for green hydrogen generation

Description

This unit outlines how to install the water feed system for an electrolyser at a green hydrogen generation facility

Scope

The scope covers the following :

- Analyse Input water system

Elements and Performance Criteria

Analyse Input water system

To be competent, the user/individual on the job must be able to:

- PC1.** Explain and illustrate the schematic of input water system for electrolyzer along with the operation of the system
- PC2.** Explain and illustrate the various processes for treating the water supply for feedstock purpose
- PC3.** Discuss various parameters essential for water as feedstock
- PC4.** Discuss and Show how the tools and tackles are utilised to set up and operate the feed water system
- PC5.** Discuss Selection of water quality for hydrogen production
- PC6.** Discuss how to monitor the quality of water feedstock for input to the electrolyzer
- PC7.** Explain and illustrate the layout of piping system for supply of water to electrolyzer
- PC8.** Explain and show how to install the Piping and Piping accessories for supplying controlled quantity of water to the electrolyzer using level controller etc.
- PC9.** Discuss and show how to record results, organize data, and perform basic computations to set up and operate input water system
- PC10.** Explain do's and don'ts in installation of water supply unit and its piping and joints

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** legislative, organizational, site requirements and related technical procedures
- KU2.** usage and handling procedure of related system components
- KU3.** tools involved in installation and maintenance activities
- KU4.** functioning of different mechanical and electrical equipment, and required precautions to handle them
- KU5.** obtain relevant authorization

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Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** complete documentation applicable to the role
- GS2.** read and correctly interpret health and safety and other technical guidelines
- GS3.** read from different sources- books, screens in machines and signage
- GS4.** read various color codes, as per standard electrical, mechanical and civil nomenclature
- GS5.** express statements with required clarity
- GS6.** make timely decisions for efficient utilization of resources
- GS7.** follow organisational code of conduct

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Analyse Input water system</i>	23	26	-	-
PC1. Explain and illustrate the schematic of input water system for electrolyzer along with the operation of the system	2	3	-	-
PC2. Explain and illustrate the various processes for treating the water supply for feedstock purpose	3	4	-	-
PC3. Discuss various parameters essential for water as feedstock	2	-	-	-
PC4. Discuss and Show how the tools and tackles are utilised to set up and operate the feed water system	2	4	-	-
PC5. Discuss Selection of water quality for hydrogen production	3	1	-	-
PC6. Discuss how to monitor the quality of water feedstock for input to the electrolyzer	3	2	-	-
PC7. Explain and illustrate the layout of piping system for supply of water to electrolyzer	2	3	-	-
PC8. Explain and show how to install the Piping and Piping accessories for supplying controlled quantity of water to the electrolyzer using level controller etc.	2	2	-	-
PC9. Discuss and show how to record results, organize data, and perform basic computations to set up and operate input water system	1	4	-	-
PC10. Explain do's and don'ts in installation of water supply unit and its piping and joints	3	3	-	-
NOS Total	23	26	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N0154
NOS Name	Analyze input water system to electrolyzer for green hydrogen generation
Sector	Green Jobs
Sub-Sector	Water Management
Occupation	Installation
NSQF Level	4
Credits	1
Version	1.0
Last Reviewed Date	NA
Next Review Date	31/01/2026
NSQC Clearance Date	31/01/2023

Qualification Pack

SGJ/N0153: Operate Hydrogen conditioning and compression unit

Description

This unit outlines how to operate the conditioning and compression unit at a green hydrogen generation facility

Scope

The scope covers the following :

- Hydrogen Conditioning
- Hydrogen Compression

Elements and Performance Criteria

Hydrogen Conditioning

To be competent, the user/individual on the job must be able to:

- PC1.** Explain the process of conditioning/purification of green hydrogen
- PC2.** Explain De-Oxo System
- PC3.** Explain Dryer System
- PC4.** Show how to undertake Installation and interconnection of major equipment for hydrogen conditioning system along with other balance of plant
- PC5.** Show how to perform selection and sizing of hydrogen tank
- PC6.** Discuss and show how to set up and operate the hydrogen conditioning system
- PC7.** Discuss other key balance of plants components

Hydrogen Compression

To be competent, the user/individual on the job must be able to:

- PC8.** Explain compression process
- PC9.** Discuss the precautions required to compress hydrogen
- PC10.** Show the functions of Green hydrogen compression system
- PC11.** Show how the compression system is set up
- PC12.** illustrate the key challenges in compression
- PC13.** Explain about H₂ compression types such as reciprocating, ionic & diaphragm.
- PC14.** Discuss the precautions required to compress hydrogen
- PC15.** Explain and Show how to perform the required safety procedures as per industry standard
- PC16.** Explain and show how to read and interpret Material Safety Data Sheet (MSDS)
- PC17.** Discuss how to select and install hydrogen compression system
- PC18.** Explain key components/ parameters of compressors including Pressure Valve , indicators, Gas Outlet & Inlets Valves, Tank size and Placement etc
- PC19.** Show how compression system comply and follow all the applicable safety and regulatory standards.

Qualification Pack

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Record keeping and filing of risk assessment, as necessary
- KU2.** Hydrogen Gas controls
- KU3.** Fault diagnosis
- KU4.** Define operating pressures for hydrogen compression
- KU5.** Correct use of notices and labels.
- KU6.** Causes and effects of pressure loss
- KU7.** Use of pressure gauge
- KU8.** The sequence of operation of hydrogen gas safety devices and controls
- KU9.** different mechanical and electrical equipment, their functioning and precautions
- KU10.** legislative, organizational, site requirements and technical procedures

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** complete documentation work applicable to the role
- GS2.** read and correctly interpret health and safety instructions, signage and guidelines
- GS3.** follow organization's rule-based decision making process
- GS4.** follow organization's code of conduct
- GS5.** make timely decisions for efficient utilization of resources
- GS6.** approach relevant authority when required

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Hydrogen Conditioning</i>	9	9	-	-
PC1. Explain the process of conditioning/purification of green hydrogen	2	2	-	-
PC2. Explain De-Oxo System	2	-	-	-
PC3. Explain Dryer System	2	-	-	-
PC4. Show how to undertake Installation and interconnection of major equipment for hydrogen conditioning system along with other balance of plant	-	2	-	-
PC5. Show how to perform selection and sizing of hydrogen tank	-	1	-	-
PC6. Discuss and show how to set up and operate the hydrogen conditioning system	1	2	-	-
PC7. Discuss other key balance of plants components	2	2	-	-
<i>Hydrogen Compression</i>	15	17	-	-
PC8. Explain compression process	2	-	-	-
PC9. Discuss the precautions required to compress hydrogen	2	-	-	-
PC10. Show the functions of Green hydrogen compression system	-	2	-	-
PC11. Show how the compression system is set up	-	3	-	-
PC12. illustrate the key challenges in compression	-	2	-	-
PC13. Explain about H ₂ compression types such as reciprocating, ionic & diaphragm.	2	-	-	-
PC14. Discuss the precautions required to compress hydrogen	2	2	-	-
PC15. Explain and Show how to perform the required safety procedures as per industry standard	1	1	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC16. Explain and show how to read and interpret Material Safety Data Sheet (MSDS)	2	2	-	-
PC17. Discuss how to select and install hydrogen compression system	2	-	-	-
PC18. Explain key components/ parameters of compressors including Pressure Valve , indicators, Gas Outlet & Inlets Valves, Tank size and Placement etc	1	2	-	-
PC19. Show how compression system comply and follow all the applicable safety and regulatory standards.	1	3	-	-
NOS Total	24	26	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N0153
NOS Name	Operate Hydrogen conditioning and compression unit
Sector	Green Jobs
Sub-Sector	Renewable Energy
Occupation	Installation
NSQF Level	4
Credits	1
Version	1.0
Last Reviewed Date	NA
Next Review Date	31/01/2026
NSQC Clearance Date	31/01/2023

Qualification Pack

SGJ/N0152: Set up and operate Hydrogen storage system

Description

This unit outlines how to set up and operate a Hydrogen storage system at a Green Hydrogen generation facility

Scope

The scope covers the following :

- Hydrogen Storage

Elements and Performance Criteria

Hydrogen Storage

To be competent, the user/individual on the job must be able to:

- PC1.** Explain the need for hydrogen storage and hydrogen storage system
- PC2.** Explain and illustrate the Challenges associated with Hydrogen in storage, handling and transportation
- PC3.** Discuss how to select and install hydrogen storage system
- PC4.** Discuss the precautions required to store hydrogen
- PC5.** Show how to perform the required safety procedures as per industry standard
- PC6.** Discuss Safety procedures – Do's & Don'ts for hydrogen storage
- PC7.** Discuss and show the safety guidelines to be followed as per applicable standards
- PC8.** Show how storage system comply and follow all the applicable safety and regulatory standards.
- PC9.** Demonstrate how to read and interpret Material Safety Data Sheet (MSDS)
- PC10.** Discuss the architecture of Piping arrangement and Basics of Storage layout requirement
- PC11.** Explain the different methods of H₂ storage (single vessel/multi cylinder cascade/portable storage/ tube-trailers etc and how these systems are set up
- PC12.** Explain about type I/II/III/IV cylinders and difference in setting up method.
- PC13.** Outline Use of special tools and tackles utilised for installing and operating hydrogen storage system
- PC14.** Discuss the Depressurization methods of Hydrogen Storage System - Use of Water seal or vent stack system
- PC15.** Explain and outline key safety measures to be followed at hydrogen compression and storage system including performing Periodic / regular leak testing of Hydrogen, Storage Systems using soap solution, carry out Gas leak detection & method of placement, understand the use of Firefighting system, Use of FRC (Flame Resistant Clothing) in Hydrogen area.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

Qualification Pack

- KU1.** statutory requirements relating to hydrogen storage
- KU2.** hydrogen specific technical standards
- KU3.** perform standard risk assessment on safety and implement measures
- KU4.** Operational maintenance requirements and procedures
- KU5.** Operational checks and procedures for storage
- KU6.** concerned Legislation, Regulations and Standards on hydrogen storage

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read from different sources- books, screens in machines and signage and correctly interpret the information
- GS2.** follow organisation's code of conduct
- GS3.** approach relevant authority when required
- GS4.** make timely decisions for efficient utilization of resources
- GS5.** read and correctly interpret health and safety guidelines and instructions
- GS6.** complete documentation work applicable to the role

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Hydrogen Storage</i>	24	26	-	-
PC1. Explain the need for hydrogen storage and hydrogen storage system	2	-	-	-
PC2. Explain and illustrate the Challenges associated with Hydrogen in storage, handling and transportation	2	3	-	-
PC3. Discuss how to select and install hydrogen storage system	2	-	-	-
PC4. Discuss the precautions required to store hydrogen	2	-	-	-
PC5. Show how to perform the required safety procedures as per industry standard	-	3	-	-
PC6. Discuss Safety procedures – Do's & Don'ts for hydrogen storage	2	1	-	-
PC7. Discuss and show the safety guidelines to be followed as per applicable standards	2	2	-	-
PC8. Show how storage system comply and follow all the applicable safety and regulatory standards.	-	3	-	-
PC9. Demonstrate how to read and interpret Material Safety Data Sheet (MSDS)	-	3	-	-
PC10. Discuss the architecture of Piping arrangement and Basics of Storage layout requirement	2	2	-	-
PC11. Explain the different methods of H ₂ storage (single vessel/multi cylinder cascade/portable storage/ tube-trailers etc and how these systems are set up	2	1	-	-
PC12. Explain about type I/II/III/IV cylinders and difference in setting up method.	2	1	-	-
PC13. Outline Use of special tools and tackles utilised for installing and operating hydrogen storage system	2	3	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC14. Discuss the Depressurization methods of Hydrogen Storage System - Use of Water seal or vent stack system	2	-	-	-
PC15. Explain and outline key safety measures to be followed at hydrogen compression and storage system including performing Periodic / regular leak testing of Hydrogen, Storage Systems using soap solution, carry out Gas leak detection & method of placement, understand the use of Firefighting system, Use of FRC (Flame Resistant Clothing) in Hydrogen area.	2	4	-	-
NOS Total	24	26	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N0152
NOS Name	Set up and operate Hydrogen storage system
Sector	Green Jobs
Sub-Sector	Renewable Energy, Other Green Jobs
Occupation	Installation
NSQF Level	4
Credits	1
Version	1.0
Last Reviewed Date	NA
Next Review Date	31/01/2026
NSQC Clearance Date	31/01/2023

Qualification Pack

SGJ/N0151: Commissioning of Green Hydrogen system

Description

This unit outlines how to commission the Green Hydrogen generation system

Scope

The scope covers the following :

- System Commissioning

Elements and Performance Criteria

System Commissioning

To be competent, the user/individual on the job must be able to:

- PC1.** Describe on the importance and use of checklists for commissioning of each hydrogen system / equipment / machinery / piping as per OEM recommendation.
- PC2.** Explain Green hydrogen system installation, testing and commissioning checklist
- PC3.** Discuss and Show how to perform various activities conforming to the checklists for commissioning of each hydrogen system / equipment / machinery / piping as per OEM recommendation
- PC4.** Discuss and show how various Inputs, outputs and key performance metrics for hydrogen generation are analysed
- PC5.** Show how to follow the checklist for Green hydrogen system installation, testing and commissioning
- PC6.** Discuss pre-commissioning tests/ Downstream considerations including hydrogen purity & pressure test, to detect leakage, Compressor Assembly operational and performance test Drying Plant Hydraulic, Safety test and performing trial run
- PC7.** Show how to arrange all instruments, equipment and facilities as required for inspection and testing at works to carry our trial run, commissioning tests
- PC8.** Explain trial run and pre-commissioning tests and the commissioning of Green Hydrogen Generation Plant in accordance with the applicable norms/ statutory requirement/ regulatory standards.
- PC9.** Discuss and Show how various Inputs, outputs and key performance metrics for hydrogen generation are analysed
- PC10.** Discuss and Show how to perform trial run and pre-commissioning tests and the commissioning of Green Hydrogen Generation Plant in accordance with the applicable norms/ statutory requirement/ regulatory standards.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** precautions to be taken while handling different electrical and mechanical components
- KU2.** legislative, organizational, site requirements and technical procedures

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- KU3.** importance of wearing protective clothing and other safety gear while carrying out maintenance activities (PPEs)
- KU4.** mechanical and electrical equipment and their functioning
- KU5.** tools involved in installation, commissioning and maintenance activities

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** complete documentation work applicable to the role
- GS2.** follow organisation's code of conduct
- GS3.** read various color codes, as per standard electrical, mechanical and civil nomenclature
- GS4.** follow organization's rule-based decision making process

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>System Commissioning</i>	24	26	-	-
PC1. Describe on the importance and use of checklists for commissioning of each hydrogen system / equipment / machinery / piping as per OEM recommendation.	4	2	-	-
PC2. Explain Green hydrogen system installation, testing and commissioning checklist	3	-	-	-
PC3. Discuss and Show how to perform various activities conforming to the checklists for commissioning of each hydrogen system / equipment / machinery / piping as per OEM recommendation	2	4	-	-
PC4. Discuss and show how various Inputs, outputs and key performance metrics for hydrogen generation are analysed	2	3	-	-
PC5. Show how to follow the checklist for Green hydrogen system installation, testing and commissioning	2	4	-	-
PC6. Discuss pre-commissioning tests/ Downstream considerations including hydrogen purity & pressure test, to detect leakage, Compressor Assembly operational and performance test Drying Plant Hydraulic, Safety test and performing trial run	3	-	-	-
PC7. Show how to arrange all instruments, equipment and facilities as required for inspection and testing at works to carry our trial run, commissioning tests	-	3	-	-
PC8. Explain trial run and pre-commissioning tests and the commissioning of Green Hydrogen Generation Plant in accordance with the applicable norms/ statutory requirement/ regulatory standards.	3	2	-	-
PC9. Discuss and Show how various Inputs, outputs and key performance metrics for hydrogen generation are analysed	3	4	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. Discuss and Show how to perform trial run and pre-commissioning tests and the commissioning of Green Hydrogen Generation Plant in accordance with the applicable norms/ statutory requirement/ regulatory standards.	2	4	-	-
NOS Total	24	26	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N0151
NOS Name	Commissioning of Green Hydrogen system
Sector	Green Jobs
Sub-Sector	Renewable Energy
Occupation	Installation
NSQF Level	4
Credits	1
Version	1.0
Last Reviewed Date	NA
Next Review Date	31/01/2026
NSQC Clearance Date	31/01/2023

Qualification Pack

SGJ/N0802: Maintain Health & Safety at Green Hydrogen generation project site

Description

This unit outlines how to perform Health and Safety measures at the Green Hydrogen generation facility

Scope

The scope covers the following :

- Safe work area

Elements and Performance Criteria

Safe work area

To be competent, the user/individual on the job must be able to:

- PC1.** Explain the requirements for safe work area at hydrogen generation project site
- PC2.** Explain the importance of Occupational health & Safety standards and regulations for Basic considerations for the safety of hydrogen systems
- PC3.** Explain the importance of administering first aid and Demonstrate how to administer first aid
- PC4.** Demonstrate the usage of personal protective equipment for ensuring safety during installation and O&M work
- PC5.** Describe potential causes of emergency such as gas leaks, fire, explosion, bomb threatening, natural calamities etc
- PC6.** Demonstrate the use of fire extinguishers, fire detection and alarm system
- PC7.** Discuss importance of different detectors and safety tools
- PC8.** Show how to comply with all applicable statutory requirements along with safety regulations in terms of fire protection.
- PC9.** Review the Material Safety Data Sheet and labels of chemicals contained in cylinders in order to be aware of their hazards and precautionary measures
- PC10.** Demonstrate how to follow necessary and adequate safety measures including personal protective equipment and precautions to avoid any accident at hydrogen generation site
- PC11.** Explain need to maintain ideal temperature and humidity level of storage areas used to safely contain gas cylinders
- PC12.** Discuss how to Utilize sensors that can alert the responsible person such as a safety officer when storage rooms are not maintaining the ideal conditions for storing hazardous chemicals
- PC13.** Identify the personal protective equipment used for the specific purpose.
- PC14.** Demonstrate how to follow necessary and adequate safety measures including personal protective equipment and precautions to avoid any accident at hydrogen generation site
- PC15.** Identify the hazards associated with hydrogen generation system
- PC16.** Identify work safety procedures and instructions for working at hydrogen generation plant complying with applicable safety regulations
- PC17.** Discuss Mock testing of fire fighting system

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- PC18.** Discuss all applicable statutory requirements along with safety regulations in terms of fire protection.
- PC19.** Show how to Incorporate good housekeeping practices and infection control guidelines.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** complete documentation work applicable to the role
- KU2.** work constructively and collaboratively with others
- KU3.** follow organization's rule-based decision making process
- KU4.** make timely decisions for efficient utilization of resources
- KU5.** read various color codes, as per standard electrical, mechanical and civil nomenclature
- KU6.** read and correctly interpret health and safety instructions and signage

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** obtain authorization from specified officials
- GS2.** legislative, organization, site requirements and technical procedures
- GS3.** usage and handling procedure related to safety of hydrogen generation equipment
- GS4.** importance of wearing protective clothing and other safety gear

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Safe work area</i>	24	26	-	-
PC1. Explain the requirements for safe work area at hydrogen generation project site	2	2	-	-
PC2. Explain the importance of Occupational health & Safety standards and regulations for Basic considerations for the safety of hydrogen systems	3	-	-	-
PC3. Explain the importance of administering first aid and Demonstrate how to administer first aid	3	-	-	-
PC4. Demonstrate the usage of personal protective equipment for ensuring safety during installation and O&M work	-	3	-	-
PC5. Describe potential causes of emergency such as gas leaks, fire, explosion, bomb threatening, natural calamities etc	2	-1	-	-
PC6. Demonstrate the use of fire extinguishers, fire detection and alarm system	-	2	-	-
PC7. Discuss importance of different detectors and safety tools	3	-	-	-
PC8. Show how to comply with all applicable statutory requirements along with safety regulations in terms of fire protection.	-	3	-	-
PC9. Review the Material Safety Data Sheet and labels of chemicals contained in cylinders in order to be aware of their hazards and precautionary measures	-	3	-	-
PC10. Demonstrate how to follow necessary and adequate safety measures including personal protective equipment and precautions to avoid any accident at hydrogen generation site	-	3	-	-
PC11. Explain need to maintain ideal temperature and humidity level of storage areas used to safely contain gas cylinders	3	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. Discuss how to Utilize sensors that can alert the responsible person such as a safety officer when storage rooms are not maintaining the ideal conditions for storing hazardous chemicals	2	-	-	-
PC13. Identify the personal protective equipment used for the specific purpose.	2	2	-	-
PC14. Demonstrate how to follow necessary and adequate safety measures including personal protective equipment and precautions to avoid any accident at hydrogen generation site	-	2	-	-
PC15. Identify the hazards associated with hydrogen generation system	-	2	-	-
PC16. Identify work safety procedures and instructions for working at hydrogen generation plant complying with applicable safety regulations	-	2	-	-
PC17. Discuss Mock testing of fire fighting system	2	-	-	-
PC18. Discuss all applicable statutory requirements along with safety regulations in terms of fire protection.	2	-	-	-
PC19. Show how to Incorporate good housekeeping practices and infection control guidelines.	-	3	-	-
NOS Total	24	26	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N0802
NOS Name	Maintain Health & Safety at Green Hydrogen generation project site
Sector	Green Jobs
Sub-Sector	Renewable Energy
Occupation	Health and Safety
NSQF Level	4
Credits	1
Version	1.0
Last Reviewed Date	NA
Next Review Date	31/01/2026
NSQC Clearance Date	31/01/2023

Qualification Pack

DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:

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- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e-mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.

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PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings

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- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	2	4	-	-
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	2	-	-
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	4	-	-
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Entrepreneurship</i>	2	3	-	-
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification will be created by the Sector Skill Council/Awarding Body. Each Element/ Performance Criteria (PC) will be assigned marks proportional to its importance in NOS.. SSC/AB will also lay down proportion of marks for Theory and Skills Practical for each Element/ PC.
2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC/AB.
3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
4. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training center (as per assessment criteria below).
5. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.
6. To pass the assessment, every trainee should score the Recommended Pass % aggregate for the Qualification.

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7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification.

Minimum Aggregate Passing % at QP Level : 70

(**Please note:** Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
SGJ/N0150.Introduce green hydrogen for clean energy transition	30	20	0	0	50	9
SGJ/N4012.Identify components of Green Hydrogen Plant and interpret its layout	30	20	0	0	50	9
SGJ/N0156.Integrate renewable power sources for Green Hydrogen Plant	24	26	0	0	50	9
SGJ/N0155.Perform installation of electrolyzer for green hydrogen generation	50	50	0	0	100	19
SGJ/N0154.Analyze input water system to electrolyzer for green hydrogen generation	23	26	0	0	49	9
SGJ/N0153.Operate Hydrogen conditioning and compression unit	24	26	0	0	50	9
SGJ/N0152.Set up and operate Hydrogen storage system	24	26	0	0	50	9
SGJ/N0151.Commissioning of Green Hydrogen system	24	26	0	0	50	9
SGJ/N0802.Maintain Health & Safety at Green Hydrogen generation project site	24	26	0	0	50	9

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National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	0	0	50	9
Total	273	276	0	0	549	100

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Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.